JAVA-PROGRAM-TO-PERFORM-ADDITION-OF-TWO-MATRICES-

AIM:

To Perform addition of two matrices using Java programming language.

APPARATUS REQUIRED:

Computer

Eclipse IDE

THEORY:

Java is a high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible. It is a general-purpose programming language intended to let programmers write once, run anywhere (WORA), meaning that compiled Java code can run on all platforms that support Java without the needto recompile. Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM) regardless of the underlying computer architecture. The syntax of Java is similar to C and C++, but has fewer low-level facilities than either of them. The Java runtime provides dynamic capabilities (such as reflection and runtime code modification) that are typically not available in traditional compiled languages. As of 2019, Java was one of the most popular programming languages in use according to GitHub, particularly for client–server web applications, with a reported 9 million developers.

Procedure for Running a Java Program in Eclipse IDE

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1. Launch Eclipse IDE

1. Open **Eclipse**.

2. Select a **workspace** (this will be the folder where your projects are saved).

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2. Create a New Project

1. Go to **File** → **New** → **Java Project**.
2. Enter the **project name**.
3. Click **Finish**.

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3. Create a New Class

1. Right-click on your **package** → **New** → **Class**.
2. Enter the **class name** (e.g., `Main`).
3. Check the box for **`public static void main(String[] args)`** (for Java).
4. Click **Finish**.

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4. Write Your Program

- Eclipse will automatically open the **code editor**.
- Type or paste your **source code**.

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5. Save the Program

- Press **`Ctrl + S`** or go to **File** → **Save**.

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6. Compile and Run the Program

1. Click the **Run** button (green ▶) on the toolbar.
2. View the output in the **Console** window.

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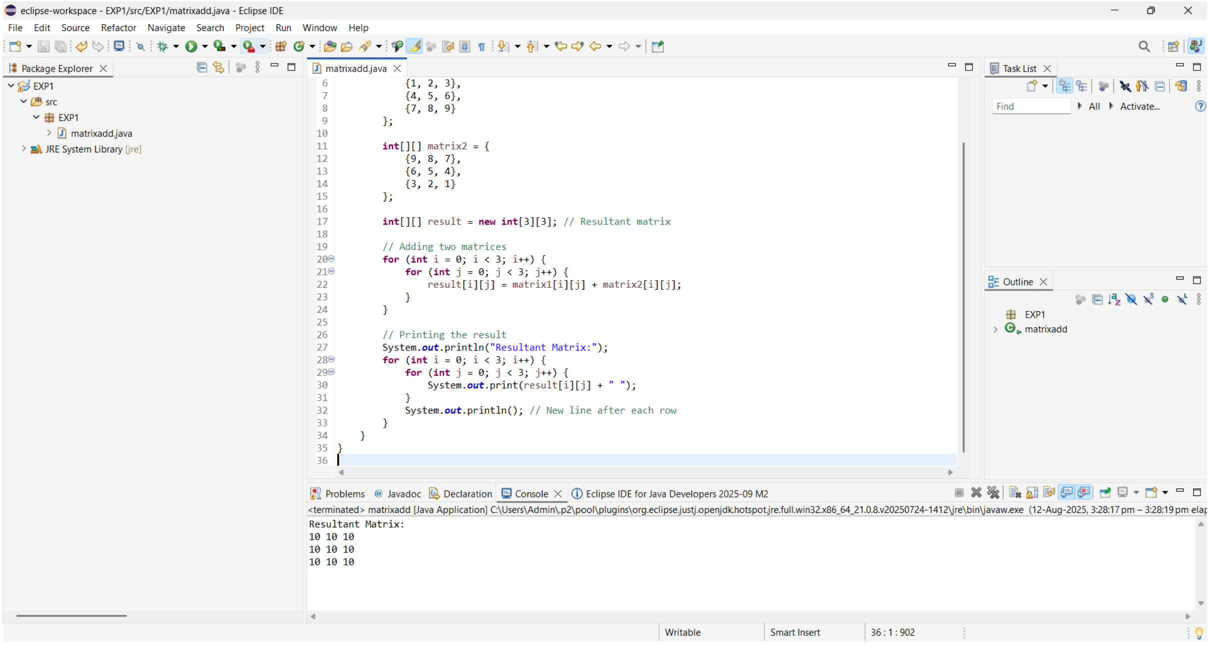
7. Close Eclipse

- After finishing, go to **File** → **Exit** to close the Eclipse IDE.

PROGRAM:

```
package EXP1;
public class matrixadd {
  public static void main(String[] args) {
     int[][] matrix1 = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
     };
     int[][] matrix2 = {
        {9, 8, 7},
        {6, 5, 4},
        {3, 2, 1}
     };
     int[][] result = new int[3][3]; // Resultant matrix
     // Adding two matrices
     for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
           result[i][j] = matrix1[i][j] + matrix2[i][j];
        }
     }
     // Printing the result
     System.out.println("Resultant Matrix:");
     for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
           System.out.print(result[i][j] + " ");
        }
        System.out.println(); // New line after each row
     }
  }
}
```

OUTPUT:



RESULT:

Thus, the program addition of two matrices using a Java program is developed, and the output is verified.